

Chapter 42

Multicam Editing

If you're working with media that was shot simultaneously using multiple cameras, then you can use the Multicam Editing tools in DaVinci Resolve to create multicam clips that can be edited using a visual switcher. Additional controls let you change the angles of multicam clips that have already been edited into the Timeline.

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Introduction to Multicam Editing

If you're working on a program where a performance, interview, or event was recorded using multiple simultaneous cameras, DaVinci Resolve has multi-camera editing tools; multicam editing for short. Editing using these tools is a three part process:

- First, you have to create multicam clips from the individual camera angles (called "ISOs," or isolated cameras).
- Second, you need to put the multicam clips you've created into a timeline.
- Third, you turn on the Multicam Viewer, and then you're ready to start cutting and switching among angles, as if you were a live multi-camera director.

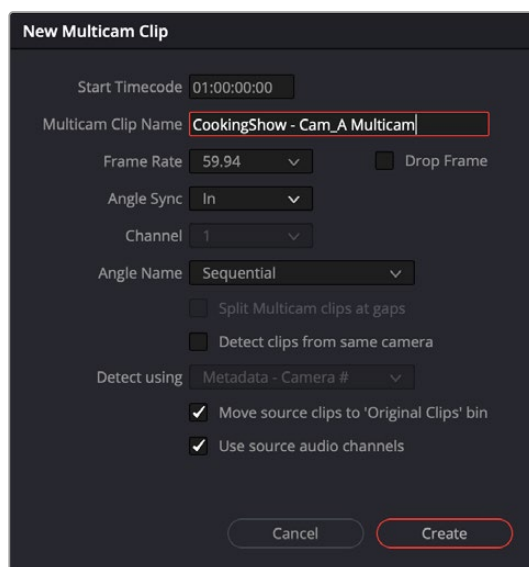
This section describes all of these steps, and the various options available for each of them.

Creating and Modifying Multicam Clips

Before you do anything else, you need to create one or more multicam clips.

To create a multicam clip:

- 1 Import all the ISO (isolated camera) clips that correspond to the multi-camera performance or event that you'll be editing into the Media Pool.
- 2 Select all the clips that you need to sync together, right-click the selection, and choose "Create Multicam Clip Using Selected Clips."
- 3 When the New Multicam Clip Properties dialog opens, choose from the following options:



New Multicam Clip Properties dialog

Start Timecode: Presents the start timecode of the new multicam clip you're about to create, which is determined by either the timecode value of the sync point if Angle Sync is defined by timecode, or by the sync point timecode value of the clip with the earliest timecode if Angle Sync is defined by waveform.

Multicam Clip Name: Use to choose a more descriptive name than "Multicam 1" for the multicam clip you're about to create.

Frame Rate: Automatically lists the frame rate associated with the clips you selected. A Drop Frame checkbox option is there if your clips are using drop frame timecode.

Angle Sync: The method used to synchronize all of the different angles. If you're manually syncing all of the angles, you can use In or Out points that you set within each clip. If matching timecode was jam synced to each camera recording an angle, you can choose Timecode for a fast sync that's as accurate as the timecode is. If each camera had a microphone with which to simultaneously record the location audio, you can choose Sound to use the shape of each audio waveform to align all of the angles. If you've manually added your own sync point markers to your clips, you can choose Marker.

Channel: If Sound is selected as your Angle Sync type, you can choose the audio channel from the clip to use as a sync source.

Angle Name: The method used to name each angle within the multicam clip being created. The angles can have Sequential numbering, use Angle or Camera metadata, or use the Clip or File name.

Split Multicam at gaps: If Sound is selected as your Angle Sync type, you can choose to split the Multicam files anywhere there is a gap in the audio.

Detect clips from same camera: Turning on this checkbox results in multiple clips that are identified as being from the same camera being put into the same angle track of the resulting multicam clip being created. It also enables the "Detect using" drop-down menu.

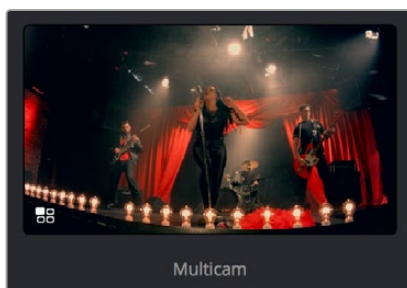
Detect Using: The metadata used to determine which clips come from the same camera. You can choose from Camera #, Angle, Reel Number, Reel Name, and Roll/Card #, which are user-editable in the Metadata Editor of the Media page, or you can choose Reel Name which is automatically or manually derived using either the Conform Options of the General Options panel of the Project Settings, or the Name panel of the Clip Attributes window.

For more information on the Conform Options, see *Chapter 4, "System and User Preferences."*

Move Source Clips to "Original Clips" Bin: A checkbox that lets you move all of the original ISO clips into an Original Clips bin to get them out of the way after the multicam clip has been created.

Use source audio channels: When checked, this gives you access to the individual tracks and channels in the source clip's audio (see below for further description).

- 4 When you're done choosing options, click Create. Depending on the Angle Sync method you chose, a waveform analysis might generate a progress bar, and then the new multicam clip is created in whichever bin is currently selected in the Media Pool. Multicam clips appear with a multicam badge in the lower left-hand corner of the clip thumbnail.

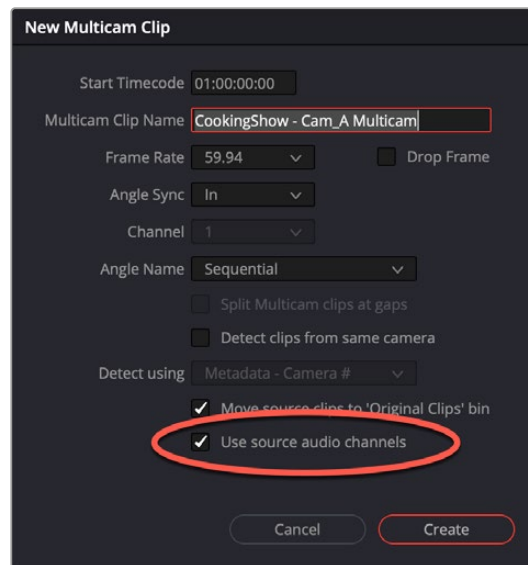


A multicam clip showing its badge in the Media Pool

Using Native Audio Channels from Source when Editing Multicam Angles

There is an option in the multicam creation dialog to Use source audio channels. This option is enabled by default. With this option enabled, the multicam clip is created in a mode where you have access to the individual tracks and channels in the source angle.

If this arrangement does not suit you, you can uncheck this option and all the audio tracks and channels for an angle are placed in a single adaptive track, wide enough to cover all the audio channels in the source. The multicam clip audio is then placed on a timeline on an adaptive track.



Using native audio channels lets you access individual tracks in the source angle.

Once the multicam clip is created with this option enabled, you can preview the audio tracks in the Audio Configuration section in the inspector File tab. With the option disabled, the multicam clip audio will be placed in an adaptive track. With the option enabled, the audio will be placed in the track format of the source.

Multicam audio clips also have the ability to select alternative mono audio channels from the timeline clip context menu (including Command right-click for faster access) and from the timeline clip audio configuration section in the Inspector.

Selecting a Single Reference Audio Track for Multicam Clips

Multicam clips in DaVinci Resolve can use reference audio files (such as those recorded by an external sound mixer) as the audio track for all angles, if you add an audio file to your multicam clip. During the creation of the multicam clip, select Reference Audio/Angle 1 in the Multicam Audio selector.

This selection automatically identifies the first audio-only angle and sets it as the reference audio for all the other angles. For multicam setups without an audio-only angle, the first angle is automatically used as the reference audio for the multicam clip. When performing multicam cuts, you can access the underlying channels in the reference audio angle by Command-right-clicking on the clip similar to source audio multicam clips.

Setting up the Reference Audio track in a multicam clip

Changing the Start Timecode for Multicam Clips

You can change the starting timecode for both multicam and compound clips. To do so, right-click on the multicam or compound clip in the Media Pool, and select Starting Timecode from the context menu. A Set New Start Timecode box will appear, then type in the new starting timecode you wish to use, and click OK.

Converting Compound Clips or Timelines to Multicam Clips

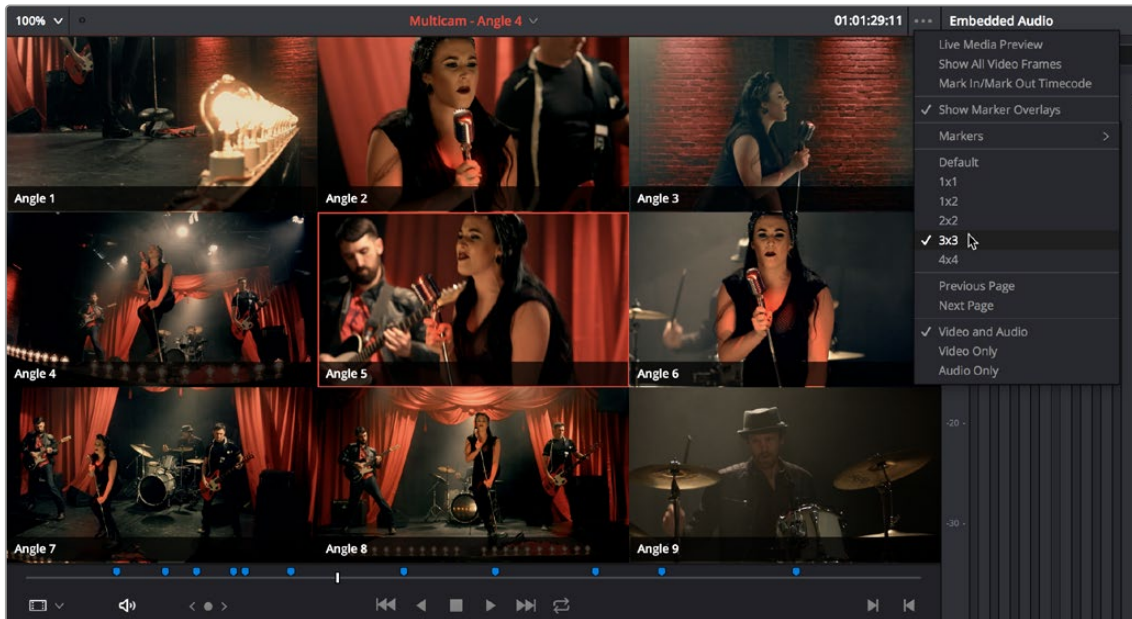
You can convert compound clips and timelines into multicam clips for easier editing using the Edit page's Multicam Editing interface. This conversion is a one-way process. You cannot reconvert a multicam clip back to a timeline or compound clip. If you wish to preserve the original timeline or compound clip, make sure to duplicate it first, and then convert the copy.

To convert a compound clip or timeline to a multicam clip:

Right-click on the clip or timeline in the Media Pool and choose "Convert Compound Clips (Timelines) to Multicam Clips" from the drop down menu.

Logging and Editing Multicam Clips

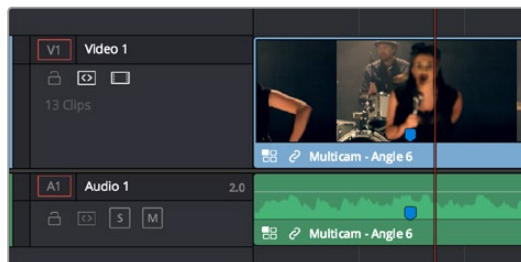
Once you create one or more multicam clips, you can view them in the Media page or in the Source Viewer of the Edit page, and add markers to them (all angles share the same markers) to prepare for the multicam edit you're planning on performing. When viewing multicam clips in the Media page, you can choose how many angles to show in the viewer via the Viewer Option menu.



Markers set in a multicam clip in the Media page to prepare for editing

Setting up a Timeline for Multicam Editing

Once you've created one or more multicam clips, preparing them for editing is as simple as editing them into the Timeline, either by dragging and dropping the multicam clip to the Timeline from the Media Pool, or by opening the multicam clip into the Source Viewer, and then using any of the available editing methods to cut it into the Timeline from there. Once edited, they appear in the Timeline like any other clip, just with a multicam badge to the left of the clip name.

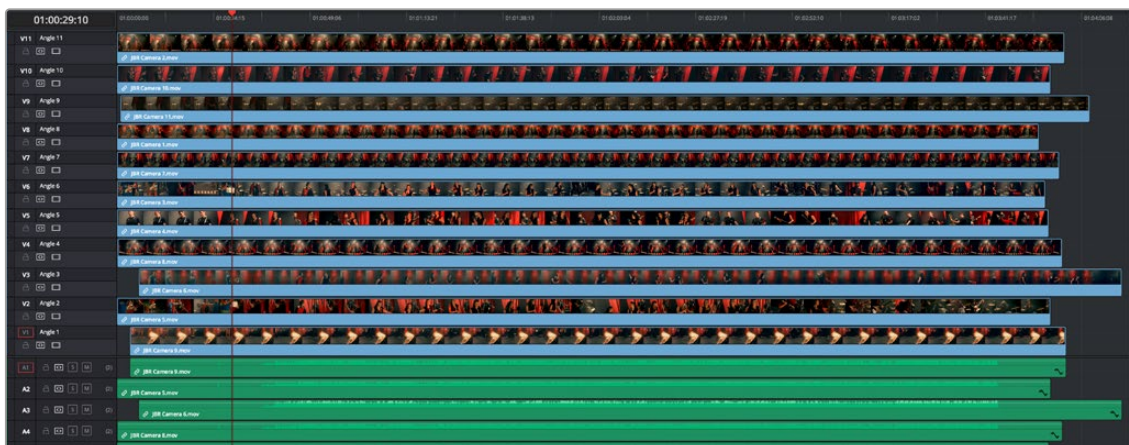


Multicam clip badge in the Timeline

When you perform a multicam edit, DaVinci Resolve plays the entire audio mix while you're editing, so if you want to take the opportunity to edit a master audio mix file or additional piece of music to play along with the multicam clip, you can do so.

Opening and Altering Multicam Clips

After you've created a multicam clip and put it into a timeline, you can modify it in a variety of ways by right-clicking it in the Media Pool and choosing "Open in Timeline." This replaces the contents of the Timeline with a vertical stack of superimposed angles, one per track, each of which is offset from the beginning of the Timeline to align with one another.

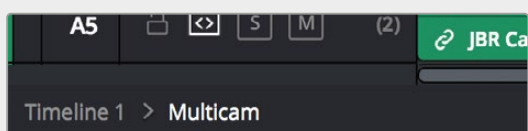


An open multicam clip appears like a timeline with a vertical stack of clips

With the multicam clip open, you can make a variety of changes in preparation for editing:

- You can slide a multicam clip left or right to alter its sync (selecting an angle and using the Period (.) and Comma (,) "nudge" keyboard shortcuts can be a good way of doing this).
- You can delete the track of an angle you don't need (right-click the track header and choose Delete Track).
- You can rearrange tracks to rearrange the order in which angles appear (right-click any track header and choose Move Track Up or Move Track Down).
- You can rename tracks to change the angle name that appears by default in the Multicam Viewer and that will also appear in the Timeline when you do cut and switch editing.
- You can disable audio or video tracks that correspond to angles you don't want to see, but don't want to eliminate, either.
- You can grade each multicam angle separately (discussed later in this section).

When you've finished altering the contents of the multicam clip, you can close it using the path control at the bottom left-hand corner of the Timeline. Click the name of the edited timeline to go back, in preparation for the next steps.

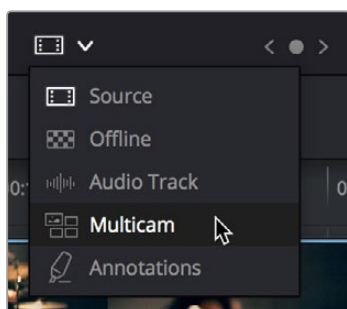


A path control lets you exit the multicam clip

Performing a Multicam Edit

After you've created one or more multicam clips and edited them into a timeline, actually executing a multicam edit is simple.

- 1 Open the Timeline you created to hold the multicam clip or clips comprising your edit, and position the playhead where you want to start editing.
- 2 Choose Multicam from the Source Viewer mode drop-down.



Switching the Source Viewer to Multicam viewing

The Source Viewer changes to display all of the different angles within that clip as switching controls.

- 3 Choose how many angles you want to display from the drop-down menu at the bottom right of the Source Viewer. If you're using a computer that's not very fast, you may need to reduce the number of angles you're viewing to maintain real time playback.

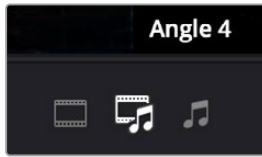


Choosing how many angles to view in the Multicam Viewer

If there are more angles within the multicam clip intersecting the playhead in the Timeline than the Multicam Viewer is set to show, then page controls appear to the left of this drop-down menu that let you choose which set of angles you want to view.

You can move to another page of angles by doing one of the following:

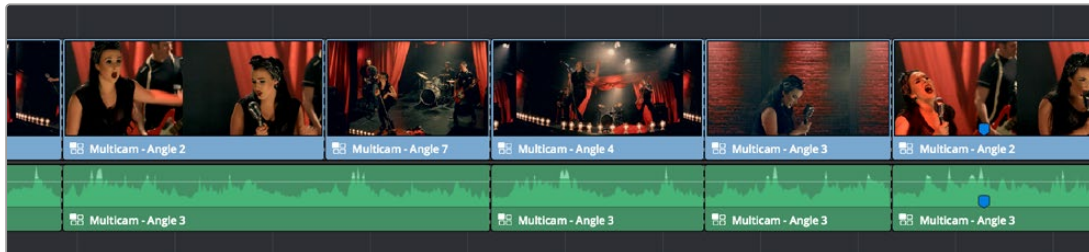
- Click any dot to jump to that page of angles.
 - Click the arrows to move among next/previous sets of angles.
 - Choose Edit > Multicam > Previous Page (Option-Shift-Left Arrow) or Next Page (Option-Shift-Right Arrow).
- 4 Choose whether you want to switch both the audio and video, just the video, or just the audio using the Audio/Video selection buttons at the bottom center of the Multicam Viewer. You can also choose Edit > Multicam > Video and Audio (Option-Shift-[), Video Only (Option-Shift-[), Audio Only (Option-Shift-]).



Buttons for choosing whether to switch the video, the audio, or both

- 5 Start playback, and while watching the program play, do one of the following:
 - Click any angle in the Multicam Viewer to insert a cut in the Timeline and switch to that angle. As you cut-and-switch, the cuts immediately appear in the Timeline while you play onward.
 - Option-click any angle to switch the angle used by the current clip without adding a cut. This is useful if you later regret the angle you cut to and just want to switch the entire segment since the last cut you made. This can also be accomplished by choosing Edit > Multicam > Previous Angle (Command-Shift-Left Arrow) or Next Angle (Command-Shift-Right Arrow).

As you play, the entire mix in the Timeline will play along with what you're switching, so you can work in context.



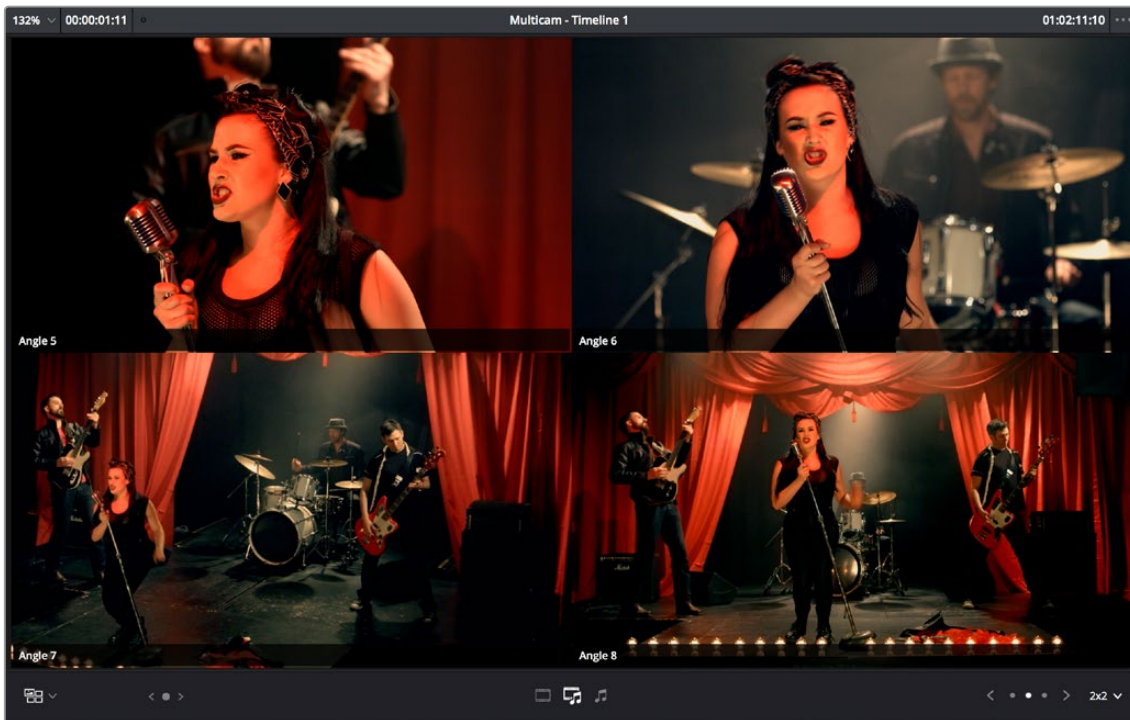
A timeline while it's being edited using cut and switch

- 6 When you're ready to stop multicam editing, simply stop playback. If you want to start trimming the Timeline to fine-tune what you've done, choose Source from the Source Viewer mode drop-down, and you can re-edit and trim the multicam clips in the Timeline just like any others.

Multicam Controls in the Source Viewer

The Source Viewer, in Multicam mode, has four sets of controls that let you set up and execute multicam editing.

- **Multicam Angle buttons:** Each multicam angle displayed in the Source Viewer is a button that lists the angle name underneath. Clicking any of these buttons inserts a cut and switches the angle of the next clip, while Option-clicking changes the angle of the clip at the position of the playhead without adding a cut.
- **Audio/Video Selection buttons:** Clicking any of these buttons toggles the audio, video, or audio and video tracks for the cut selection of the Multicam clip. Audio makes an audio only cut, video makes a video only cut, and audio and video cuts both.
- **Multicam display drop-down:** Lets you choose how many angles to view while switching. Depending on your workstation's performance, reducing the number of angles can improve playback performance while you edit. You can choose from a grid of 1x1, 1x2, 2x2, 3x3, or 4x4 angles to view.
- **Multicam Page buttons:** If there are more angles within the multicam clip intersecting the playhead in the Timeline than the Multicam Viewer is set to show (via the multicam display drop-down), then page controls appear that let you choose which set of angles you want to view. Click any dot to jump to that page of angles, or click the arrows to move among next/previous sets of angles.



The Source Viewer showing Multicam switching controls

Multicam Keyboard Controls

There's also a full set of keyboard shortcuts that can be used for multicam editing.

- **Multicam Cut:** (Clip > Multicam Cut submenu) Pressing the 1 through 9 number keys performs a cut-and-switch operation, the same as if you'd clicked on an angle button of a multicam clip in the Source Viewer.
- **Multicam Switch:** (Clip > Multicam Switch submenu) Pressing Option-1 through 9 performs a switch operation, the same as if you'd Option-clicked an angle button of a multicam clip in the Source Viewer.
- **Previous/Next Angle:** (Edit > Multicam submenu) Pressing Command-Shift-Left or Right Arrow lets you switch to the previous or next angle. These controls will also loop back around to the first or last angles in the multicam clip.
- **Audio/Video Switching:** (Edit > Multicam submenu) Pressing Option-Shift-[sets the Multicam Viewer to cut or switch both Video and Audio at the same time. Pressing Option-Shift-] sets the Multicam Viewer to cut or switch Video only. Pressing Option-Shift-\ sets the Multicam Viewer to cut or switch Audio only.
- **Previous/Next Page:** (Edit > Multicam submenu) Pressing Option-Shift-Left or Right Arrow lets you move to the previous or next page of multicam angles, if there are more angles than can be displayed in the Viewer's current multi-angle setting.

Editing Multicam Clips in the Timeline

When it comes to editing and trimming, there's no functional difference between multicam clips and any other kind of clip. Because you're technically adding through edits to a single clip, you have the option of deleting any edit by selecting it and pressing the Delete key.

But multicam clips are special in that you always have the option of switching angles, either using the Multicam Viewer, or right in the Timeline via each clip's contextual menu.

To switch the angle of any multicam clip in the Timeline:

- Right-click any clip and choose a new angle from the "Switch Multicam Clip Angle" submenu. This also allows you to change angles without needing to use the Multicam Viewer.

In the event that you want to eliminate all unused angles from a multicam clip and "flatten" it to simply be a single clip in the Timeline, there's a command for that.

To flatten a multicam clip in the Timeline:

- Right-click any clip and choose Flatten Multicam Clip from the contextual menu. Select either Copying Multicam Grades to apply the existing color grade to the flattened multicam angles, or Retaining Grades from Angles to keep any color grades done on individual angles. All unused angles are deleted, the clip becomes shorter if it included black tails because of another unused angle, and you end up with a single ordinary clip in the Timeline.

To Match Frame to the angle of any multicam clip in the Timeline:

- Place the playhead over the frame in the Timeline you want to match and press F. The exact frame of the multicam clip referenced will appear in the Source Viewer with the appropriate angle already selected.

To Match Frame to the multicam clip in the Source Viewer:

- Open the multicam clip in the Source Viewer. Navigate to the frame you want to find using the jog bar and press F. The playhead in the Timeline will move to the exact frame of the multicam clip referenced in the Source Viewer. If the frame you selected is not in the Timeline, when you press F nothing will happen.

To edit an angle of any multicam clip directly from the Source Viewer:

- You can click and drag any multicam angle directly from the Source Viewer to the Timeline. The length of the clip is bounded by the In and Out point selection of the clip.

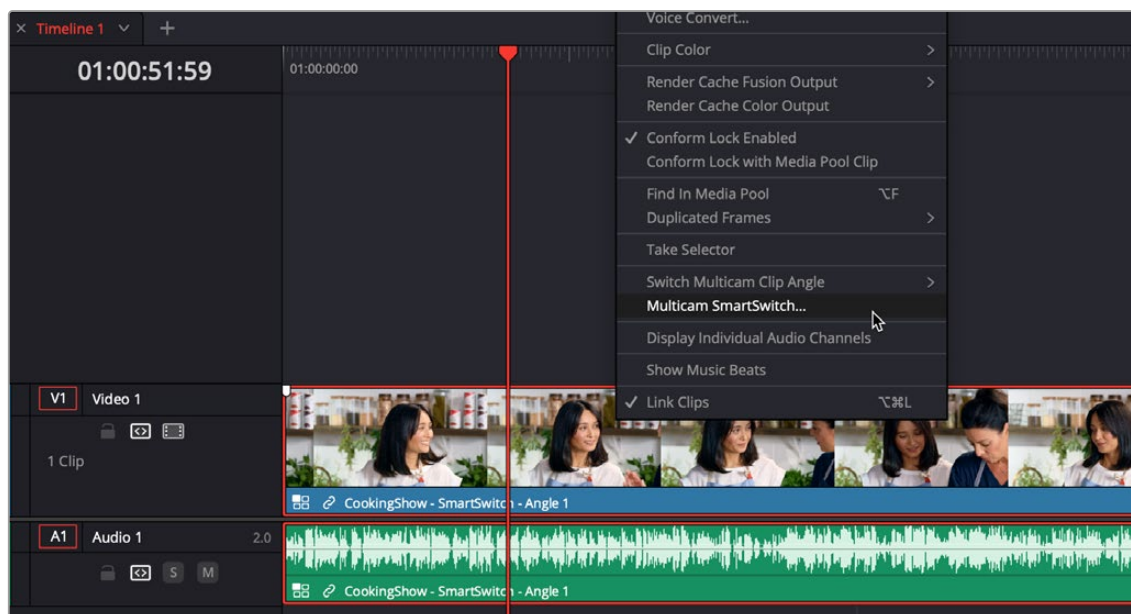
Referencing a Line Cut

You may sometimes be provided with what's called a "line cut" from a production. This is a pre-edited version of the program, cut live with the switcher and recorded during the performance or event, that's meant to be used as a reference for what you're doing. If you want to reference a line cut that's been given to you as a movie file, you can add it as an Offline Reference Movie, and compare it to the Timeline using the Offline Reference Movie mode of the Source Viewer in the Edit page.

For more information on using an Offline video to compare with a timeline in the Edit page, see *Chapter 55, "Preparing Timelines for Import and Comparison."*

Multicam SmartSwitch (Studio Version Only)

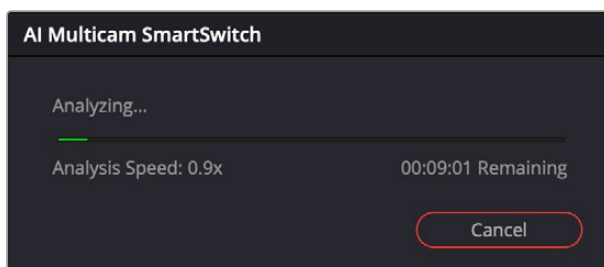
DaVinci Resolve 20 has a new AI-powered multicam tool called Multicam SmartSwitch, which will analyze all multicam camera angles and automatically cut to the most appropriate angle, based on who is the active speaker. Multicam SmartSwitch doesn't just use the audio track to determine which angle is best but includes other video related traits, such as lip movement in the frame, and if the shot is a wide or close up. Trained on thousands of hours of multicam footage, Multicam SmartSwitch is great for giving you a good first pass and letting you finesse the rest.



Selecting a multicam clip to be used with Multicam SmartSwitch

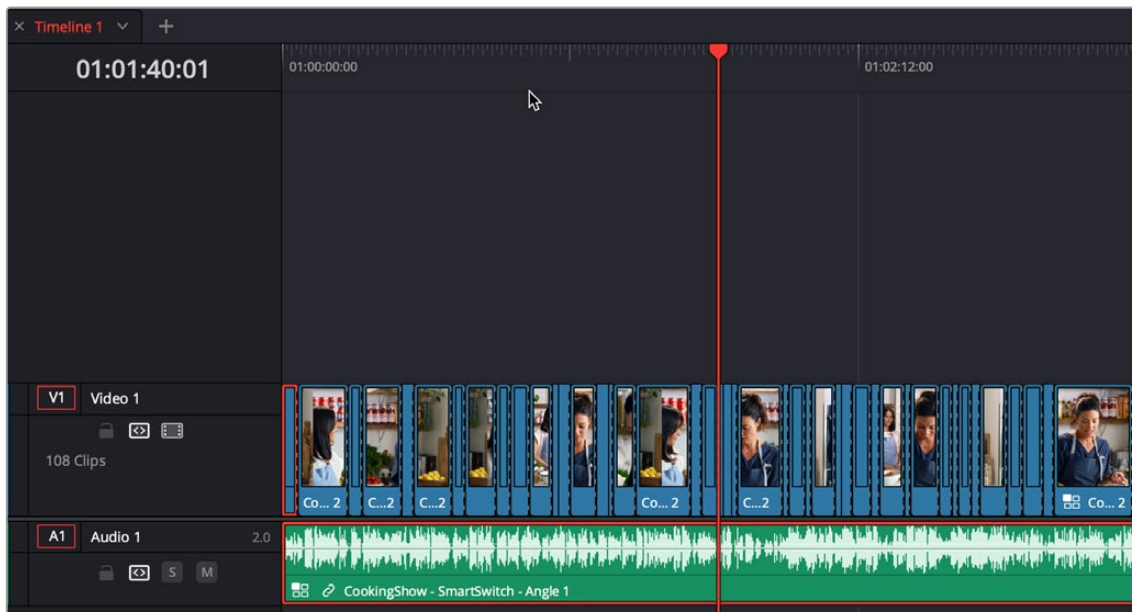
To use Multicam SmartSwitch to automatically cut your multicam clip:

- 1 Create a multicam clip in the Media Pool of all the individual camera angle clips, as described previously in this chapter.
- 2 Edit the multicam clip into the timeline.
- 3 Right-click on the multicam clip, and select Multicam SmartSwitch from the context menu.
- 4 Select the appropriate settings and parameters from the Multicam SmartSwitch tool.
- 5 Click Analyze.



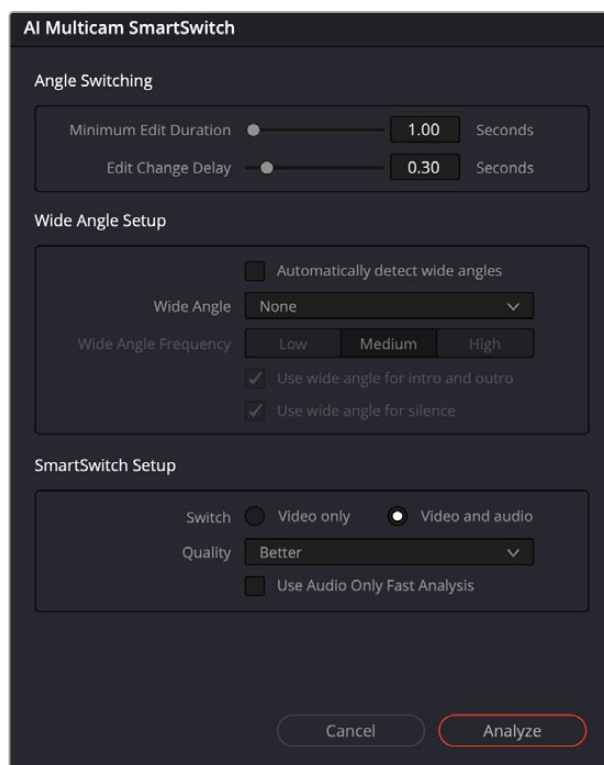
Multicam SmartSwitch analyzing the multicam clip

At this point, Multicam SmartSwitch will analyze all your video angles and audio tracks and choose what it thinks the best angle is under the circumstances. This can take quite some time, depending on the length and number of cameras angles used in the multicam shoot.



Multicam SmartSwitch finished cut on the timeline

When it is finished, you will have a fully cut version of your multicam clip available in the timeline. If you don't like a choice that Multicam SmartSwitch made, you can easily switch the camera angle by right-clicking on it and selecting a new one from the Switch Multicam Clip Angle context menu, or using the standard trim tools to change the edit point.



The Multicam SmartSwitch tool

Multicam SmartSwitch Settings

You can set parameters for the Multicam SmartSwitch tool that modifies how it makes its cut decisions.

Angle Switching: Adjusts the timing of the edits.

- **Minimum Edit Duration:** Sets the minimum length of an edit. Regardless of what happens in the cut, Multicam SmartSwitch won't change angles before the end of this duration.
- **Edit Change Delay:** This lets you set the amount of time between the person speaking and the edit changing.

Wide Angle Setup: Just like a technical director, Multicam SmartSwitch will tend to use wide angles to cover things like people talking over each other and silences. You can tweak those settings here.

- **Automatically detect wide angles:** Check this box to have Multicam SmartSwitch automatically choose what it considers to be wide shots.
- **Wide Angle:** If you have a specific wide shot you want to use for coverage, you can manually select it here.
- **Wide Angle Frequency:** How often should Multicam SmartSwitch cut back to the wide shot. Choices are Low, Medium, and High.
- **Use wide angle for intro and outro:** Check this box to open and close the multicam clip on a wide shot.
- **Use wide angle for silence:** Check this box to go back to the wide shot when no one is speaking.

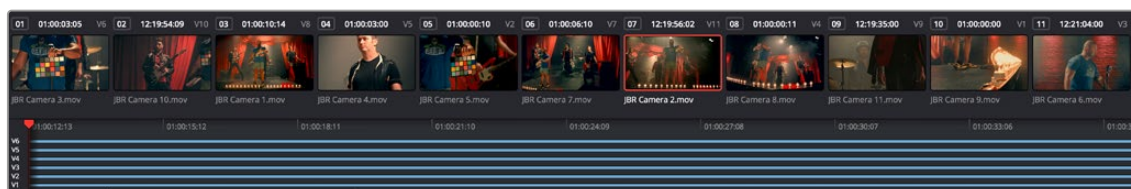
SmartSwitch Setup: Determines what media types to use for the analysis.

- **Switch:** You can choose for the cuts to be Video Only or cut Video and Audio tracks together.
- **Use Audio Only Fast Analysis:** Check this box to use an analysis that uses only the audio to make its decisions, rather than analyze the video as well. This runs much faster and is only available in certain modes.

Grading Multicam Clips

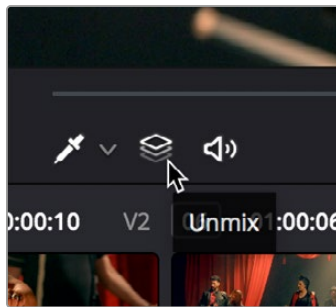
Multicam clips appear like any other clip in the Color page. However, each angle within a multicam clip has its own grade (unlike the Take Selector described later, in which all takes share the same grade). If you grade a multicam clip, you're actually editing the grade of the specific angle that's currently exposed in that clip.

If you want access all of the angles within a multicam clip for grading, right-click it and choose "Open in Timeline" to expose each angle within a superimposed stack. Then you can open the Color page and grade whatever angles you want, whether they're visible back in the Edit page or not. You might do this to make the different angles match one another better, or to pre-grade all of the angles to give them the look you want prior to multicam editing. You also need to use "Open in Timeline" in order to access and manipulate the camera RAW settings if your footage was shot in a RAW format.



An open multicam clip in the Color page exposes all of its angles for individual grading

Because opening multicam clips in the Timeline results in a vertical stack of superimposed clips, you'll want to turn on Unmix in the Color page viewer so that you can actually see the currently selected angle in the Thumbnail Timeline while you work.



The Unmix control lets you see only one of a superimposed stack of clips

When flattening a multicam clip, you can choose to grade individual angles or the multicam clip as a whole. In the Edit page, select Flatten Multicam Clip and either Copying Multicam Grades to apply the existing color grade to the flattened multicam angles, or Retaining Grades from Angles to keep any color grades done on individual angles.

When you're done grading, go back to the Edit page, and use the path control at the bottom left-hand corner of the Timeline to return to your edited Multicam timeline.